

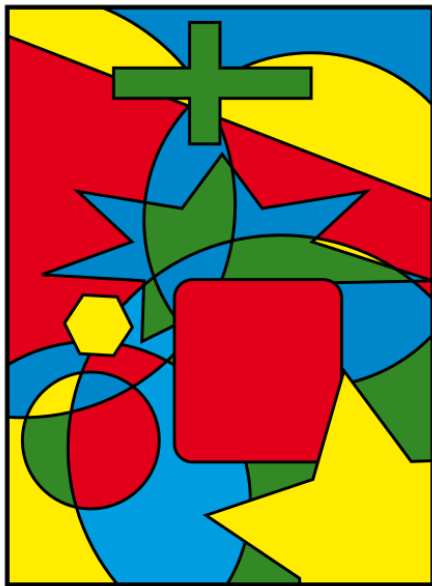
10-planarity-coloring: 平面图与图着色 (Planarity and Coloring)

魏恒峰

hfwei@hnu.edu.cn

2026 年 05 月 21 日



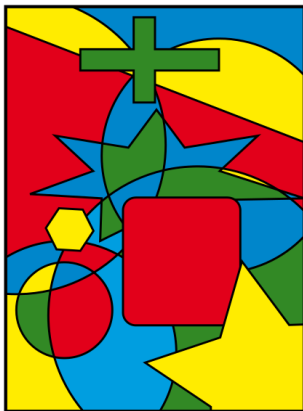


Theorem (Four Color (Map) Theorem (informal))

*Every **map** can be colored with only **four** colors such that no two **adjacent regions** share the same color.*

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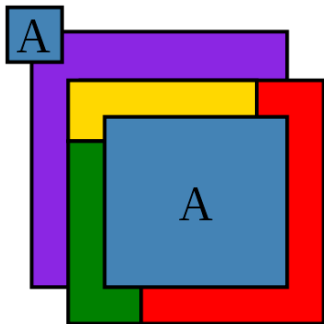


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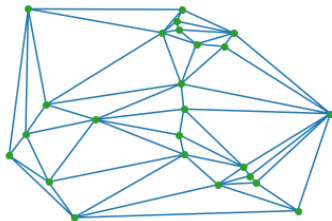
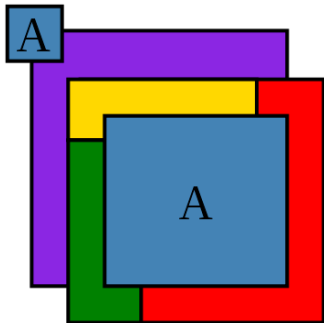
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Regions should be **contiguous**.

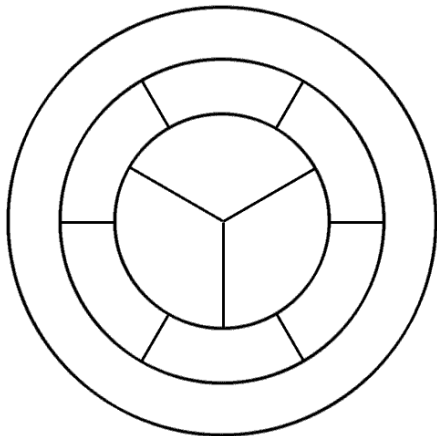
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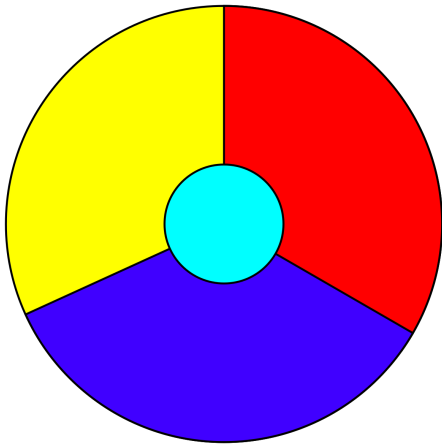
Adjacent regions share a segment.

Regions should be **contiguous**.



4 colors are necessary

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Every region is adjacent to the other 3 regions.

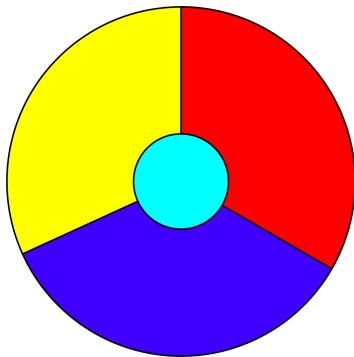
What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

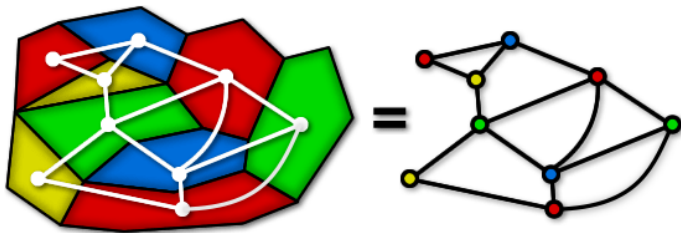
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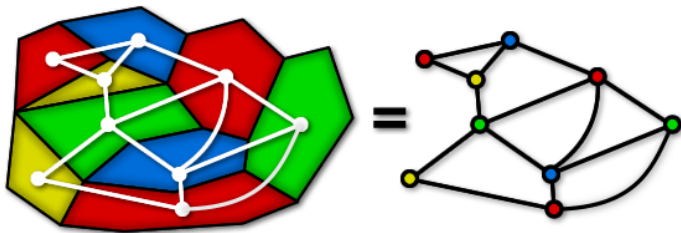


IMPOSSIBLE™

What does Four Color Theorem to do with Graph Theory?



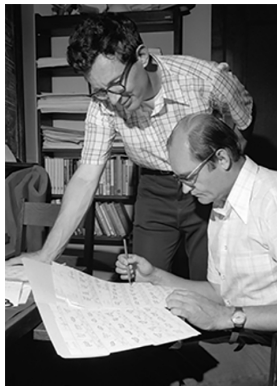




Every map produces a **planar** graph.

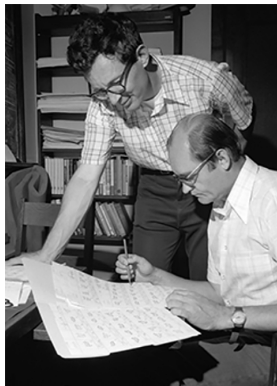
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I will *not* show its proof (which I don't understand either)!

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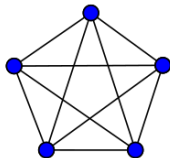
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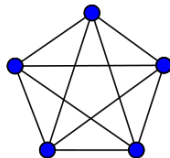
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What if we have a map which contains *5* regions so that every region is adjacent to the other *4* regions?

Theorem

K_5 is not *planar*.

Theorem

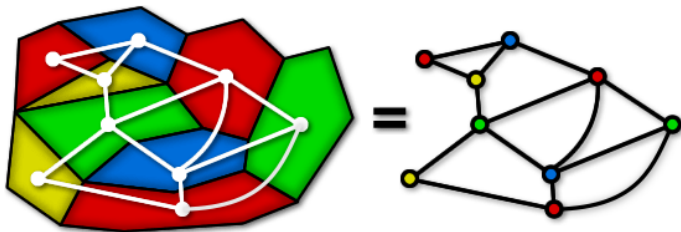
Every simple planar graph is 6-colorable.

Theorem

Every *simple planar* graph is *6-colorable*.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.



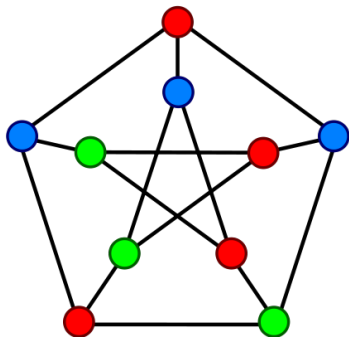
Graph Coloring Problem

Definition (k -Colorable (k -可着色的))

If G is a connected undirected graph without loops, then G is **k -colorable** if its vertices can be colored in k colors so that adjacent vertices have different colors.

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The Petersen graph is ≥ 3 -colorable.

Definition (k -Chromatic (k -色数的))

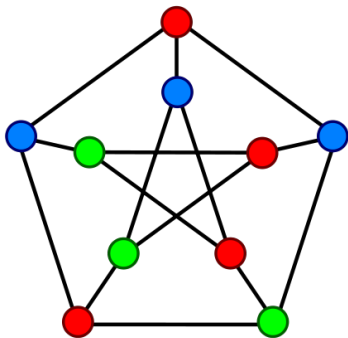
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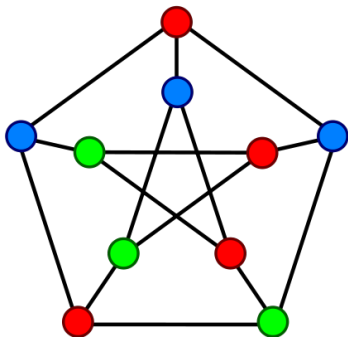


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The Petersen graph is **3**-chromatic.
(It contains an **odd** cycle.)

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A graph is 1-colorable iff

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A graph is 1-colorable iff it is the empty graph with 1 vertex.

Theorem

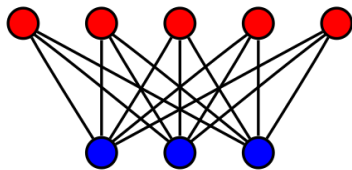
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Theorem

*A graph is 2-colorable iff it is **bipartite**.*

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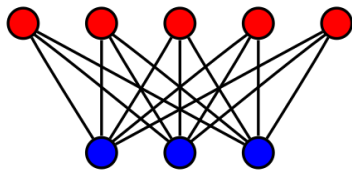
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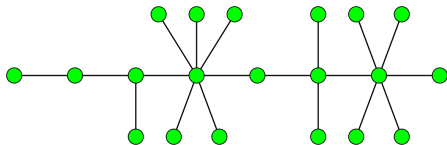
$K_{5,3}$

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$K_{5,3}$



Trees are bipartite.

Theorem (Characterization of Bipartite Graphs)

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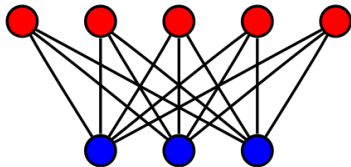
*A graph is 2-colorable iff it is **bipartite**.*

*A graph is **bipartite** iff it does not contain any **odd** cycles.*

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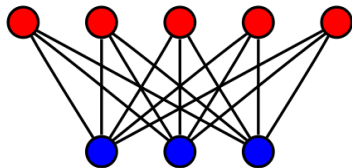


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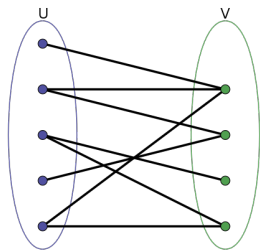
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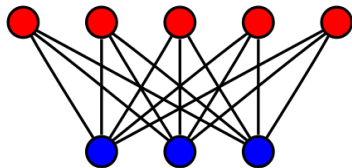
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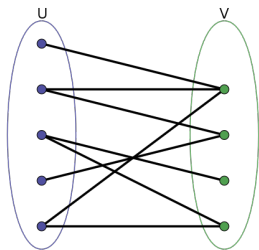
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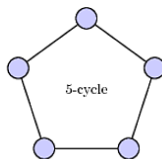
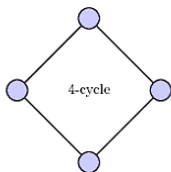
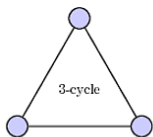


Lemma

Every tree is bipartite and is thus 2-colorable.

Lemma (Characterization of Bipartite Graphs ($\implies$))

If a graph is *bipartite*, then it does not contain any *odd* cycles.



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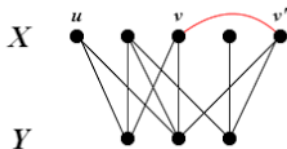
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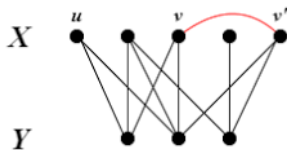
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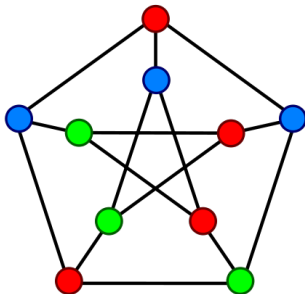
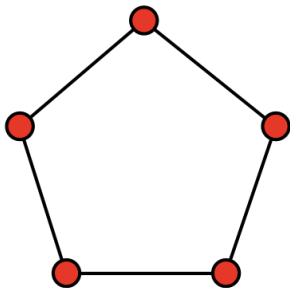
$u \rightsquigarrow v \rightarrow v' \rightsquigarrow u$ is an **odd** cycle.

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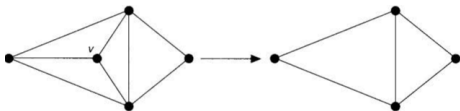
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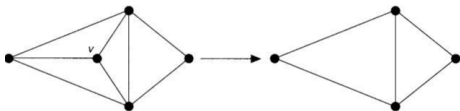
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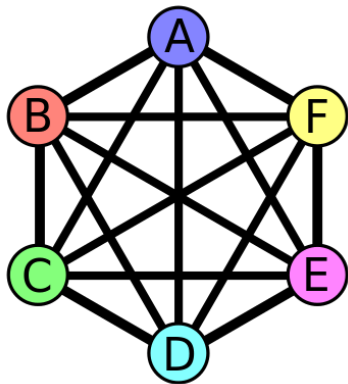
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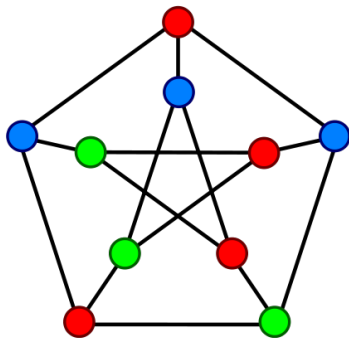
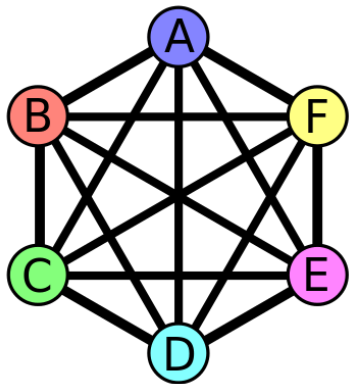
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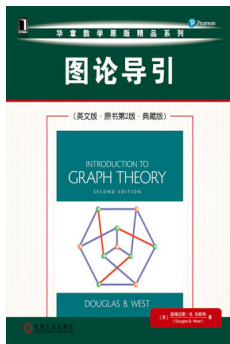
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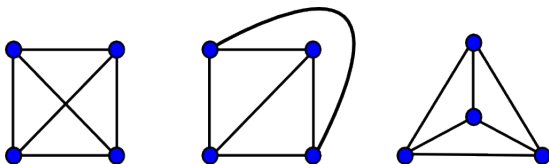
Theorem 5.1.22

Definition (Planar Graph (平面图))

A **planar graph** is a graph that **can** be drawn in the plane without **edge crossings**.

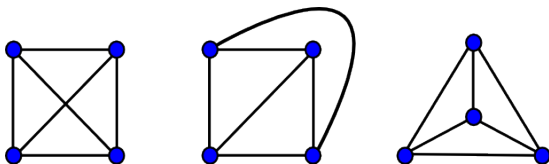
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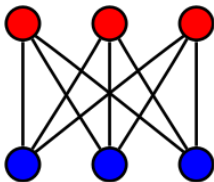


Theorem (K. Wagner (1936); I. Fáry (1948))

*Every simple planar graph can be drawn with **straight lines**.*

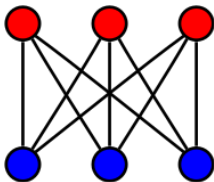
Theorem (Kazimierz Kuratowski, 1930)

The utility graph $K_{3,3}$ is non-planar.



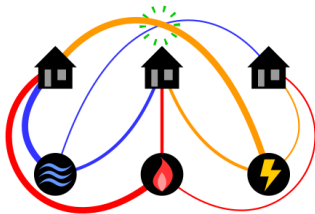
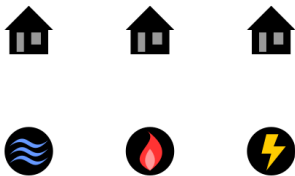
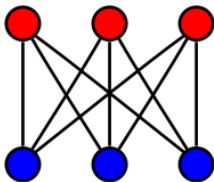
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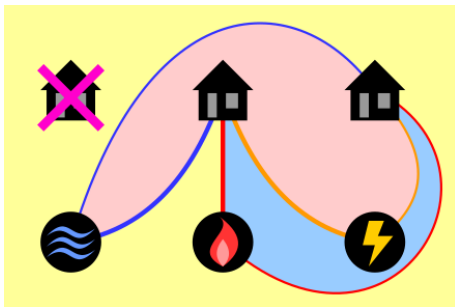
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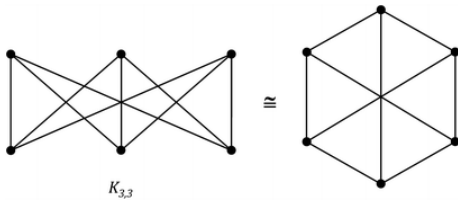
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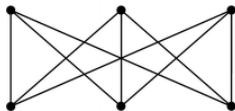
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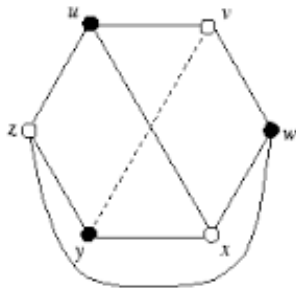
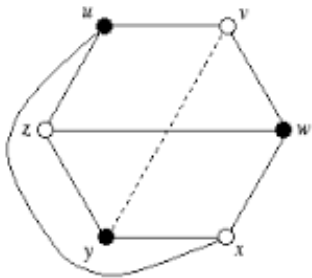
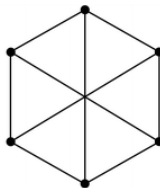
Proof without Words

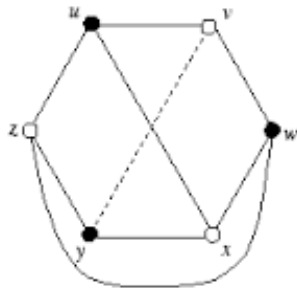
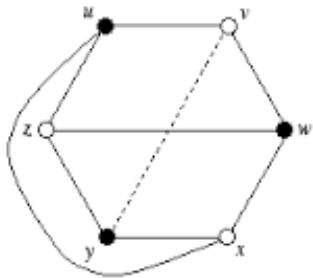
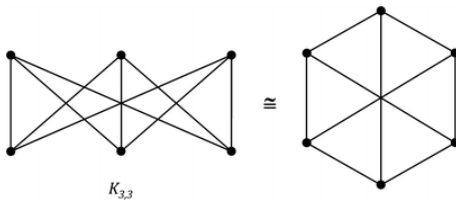




$K_{3,3}$

$\cong$

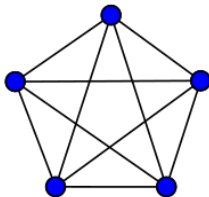




$$\text{cr}(K_{3,3}) = 1 \quad (\text{crossing number})$$

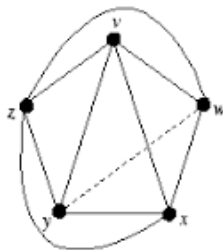
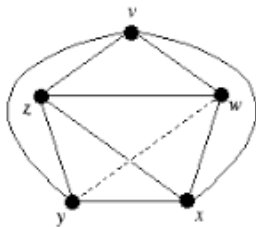
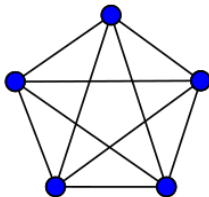
Theorem

K_5 is non-planar.



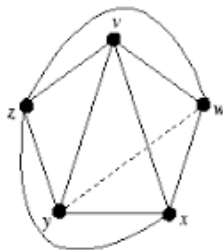
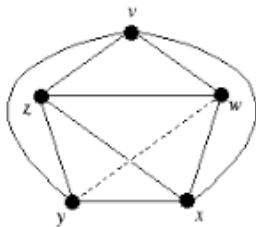
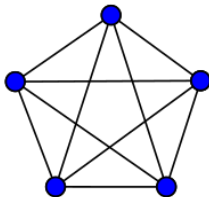
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Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.

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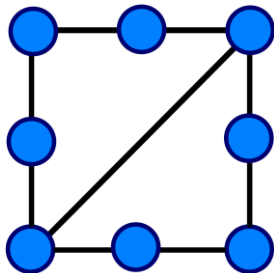
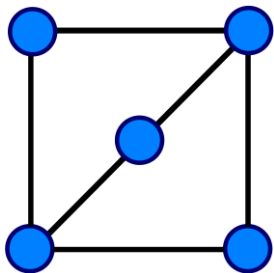
*“The K in K_5 stands for Kazimierz,
and the K in $K_{3,3}$ stands for Kuratowski.”*

Definition (Homeomorphic)

Two graphs are **homeomorphic** if one can be obtained from another by **inserting or contracting** vertices of **degree 2**.

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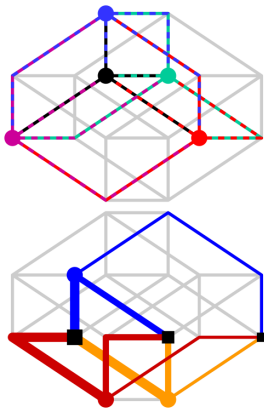


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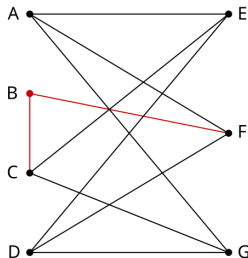
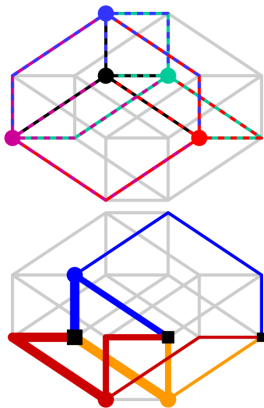
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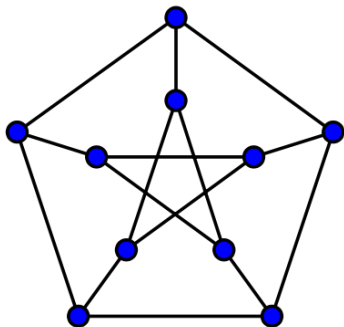
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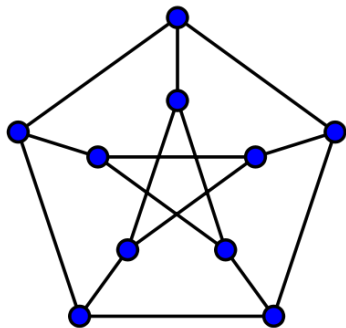
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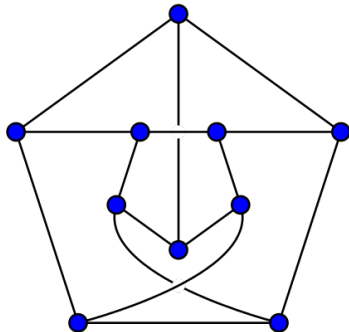
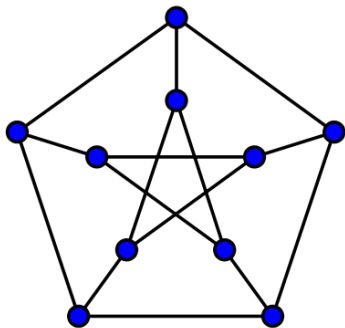
The Petersen graph is non-planar.

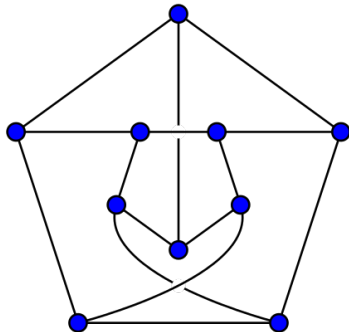
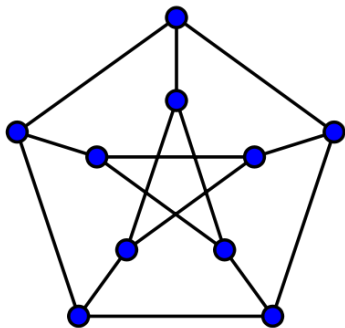


[https:](https://github.com/courses-at-hnu-by-hfwei/math4cs-lectures/blob/main/10-planarity-coloring/figs/Kuratowski-Petersen.gif)

[//github.com/courses-at-hnu-by-hfwei/math4cs-lectures/blob/main/10-planarity-coloring/figs/Kuratowski-Petersen.gif](https://github.com/courses-at-hnu-by-hfwei/math4cs-lectures/blob/main/10-planarity-coloring/figs/Kuratowski-Petersen.gif)







$$\text{cr}(\text{Petersen Graph}) = 2$$

Efficient Planarity Testing

JOHN HOPCROFT AND ROBERT TARJAN

Cornell University, Ithaca, New York

ABSTRACT. This paper describes an efficient algorithm to determine whether an arbitrary graph G can be embedded in the plane. The algorithm may be viewed as an iterative version of a method originally proposed by Auslander and Parter and correctly formulated by Goldstein. The algorithm uses depth-first search and has $O(V)$ time and space bounds, where V is the number of vertices in G . An ALGOL implementation of the algorithm successfully tested graphs with as many as 900 vertices in less than 12 seconds.

KEY WORDS AND PHRASES: algorithm, complexity, depth-first search, embedding, genus, graph, palm tree, planarity, spanning tree

CR CATEGORIES: 3.24, 5.25, 5.32



A planar graph should not has too many edges.

Theorem (Euler's Formula, 1750)

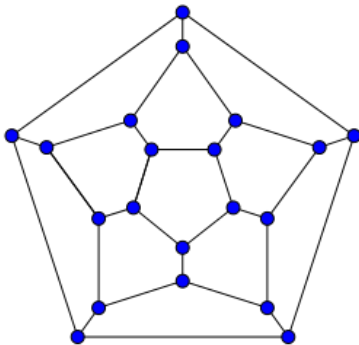
Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

$$n - m + f = 2$$

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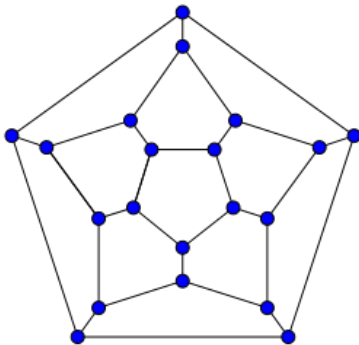
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$$n - m + f = 20 - 30 + 12 = 2$$

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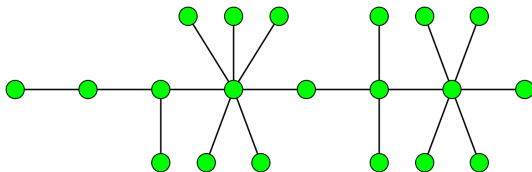
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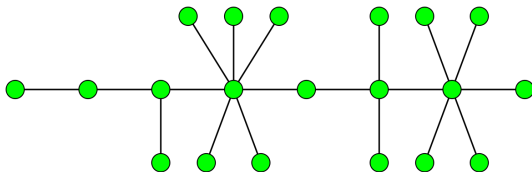
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Therefore,

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Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. Then

$$m \leq 3n - 6.$$

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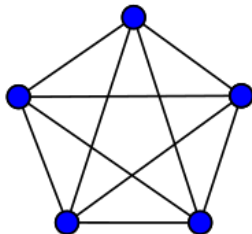
Double Counting the Edges around Faces:

each edge bounds 2 faces;

each face is bounded by ≥ 3 edges

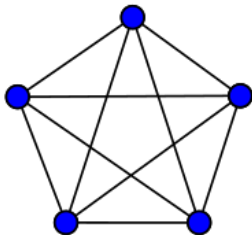
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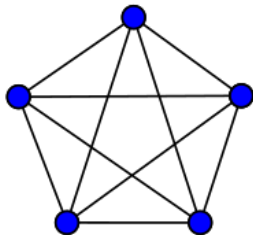
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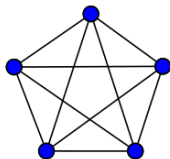
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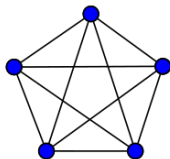


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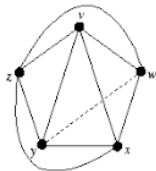
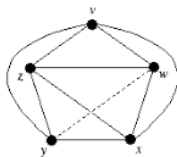
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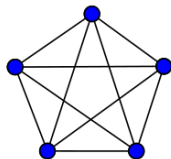
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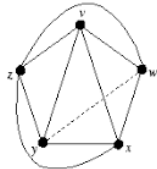
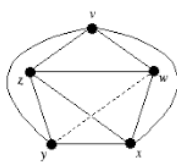
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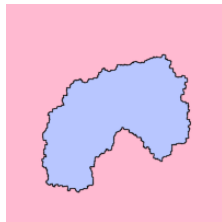


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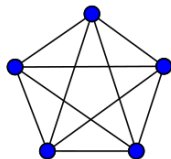


Theorem (Jordan Curve Theorem (1887))



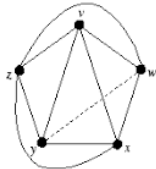
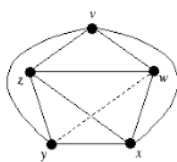
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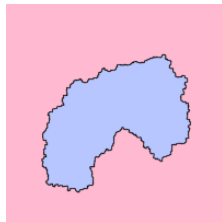


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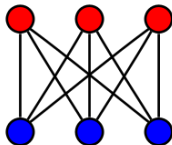


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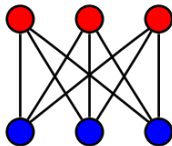
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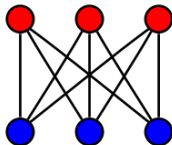
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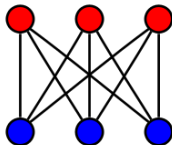


$$m \leq 3n - 6$$

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FAILED

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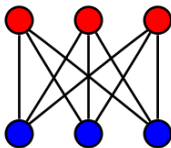
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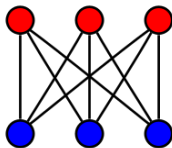
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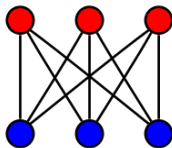
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$G' = G - v$ is *5-colorable*.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

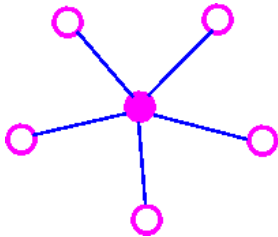
Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

$G' = G - v$ is *5-colorable*.

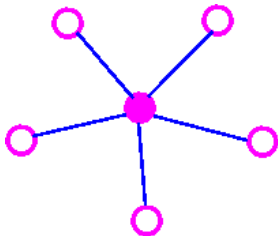
If $\deg(v) < 5$, G is 5-colorable.

Now assume that $\deg(v) = 5$.



$$\{v, v_1\}, \{v, v_2\}, \{v, v_3\}, \{v, v_4\}, \{v, v_5\}$$

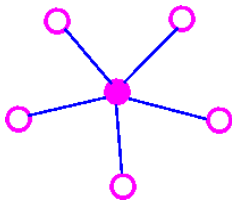
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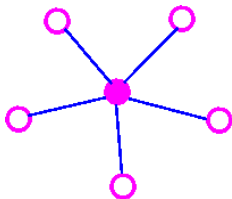
$$\{v, v_1\}, \{v, v_2\}, \{v, v_3\}, \{v, v_4\}, \{v, v_5\}$$

If v_1, v_2, v_3, v_4 , and v_5 uses < 5 colors, we are done.

Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.

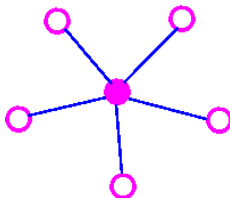


Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.

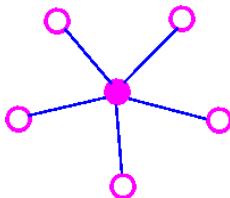
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Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored **red** or **blue**.

Let S be the set of all **red** or **blue** vertices of $G - v$
connected to v_1 by a **red-blue** path.

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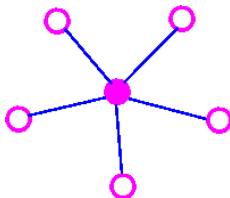


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$$v_1 \in S, \quad v_3 \notin S$$

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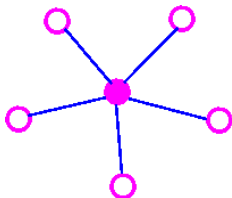
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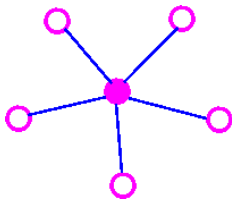
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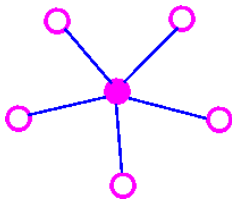
Interchange the colors of the **red** or **blue** vertices in S

Coloring v **red** produces a 5-coloring of G .

Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.

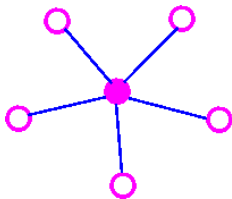


Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
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There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

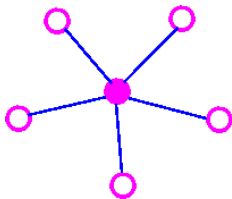
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By similar argument, G is 5-colorable.

$P(G, k)$: How many ways to color the vertices of G with k colors?

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Theorem (Deletion-Contraction Recurrence)

Let G be a simple graph and e be an edge of G . Then

$$P(G, k) = P(G - e, k) - P(G/e, k),$$

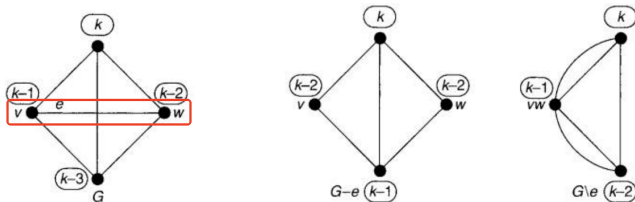
where $G - e$ is the graph obtained from G by *deleting* e ,
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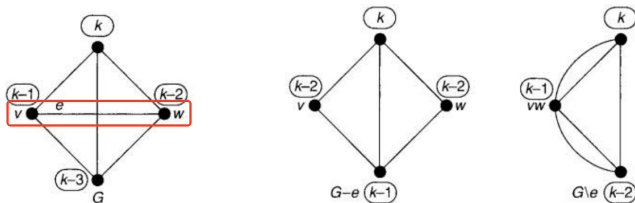


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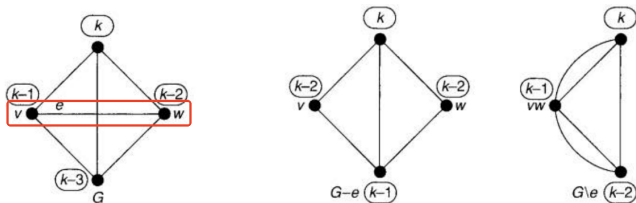
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$$k(k-1)(k-2)(k-3) = k(k-1)(k-2)^2 - k(k-1)(k-2)$$

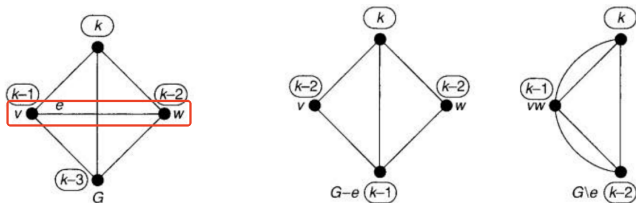
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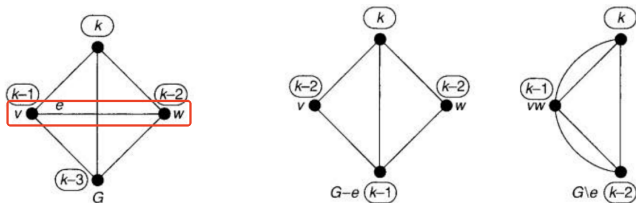
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$$P(G - e, k) = P(G, k) + P(G/e, k)$$

Theorem (Deletion-Contraction Recurrence)

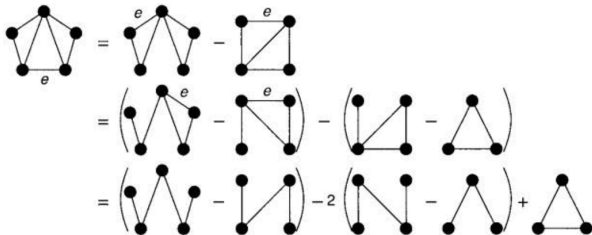
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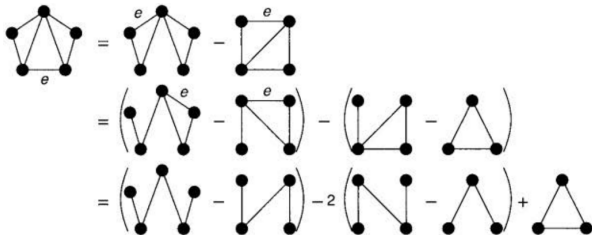
$$P(G - e, k) = P(G, k) + P(G/e, k)$$

$P(G - e, k) = P(G - e, k)$ with v and w having *different* colors
 + $P(G - e, k)$ with v and w having the *same* color

$$P(G, k) = P(G - e, k) - P(G/e, k)$$

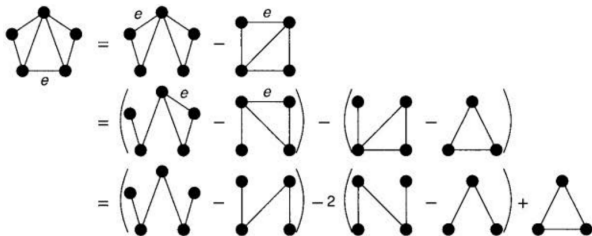


$$P(G, k) = P(G - e, k) - P(G/e, k)$$



$$\begin{aligned} P(G, k) &= k(k-1)^4 - 3k(k-1)^3 + 2k(k-1)^2 + k(k-1)(k-2) \\ &= k^5 - 7k^4 + 18k^3 - 20k^2 + 8k \end{aligned}$$

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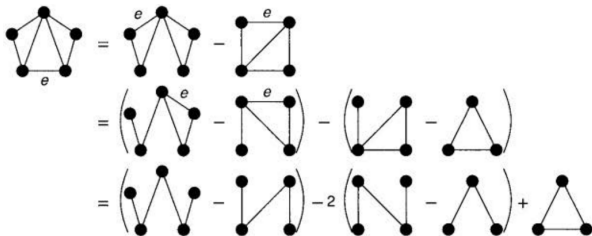


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Theorem (Chromatic Polynomial)

$P(G, k)$ is a polynomial in k .

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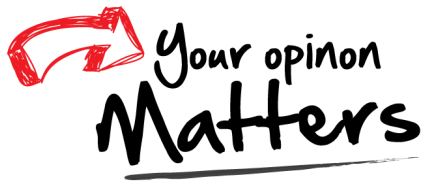
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Theorem (Chromatic Polynomial)

$P(G, k)$ is a polynomial in k .

$$k^n - mk^{n-1} + \dots - \dots + ak + 0$$

Thank
You!



hfwei@hnu.edu.cn